CAPSTONE PROJECT PHASE -2

UE23CS320B - PHASE 2 ESA REVIEW

Project Title: Federated Learning Architecture for Health Monitoring and Anomaly Detection in Multi Satellite Systems

Project ID: 122

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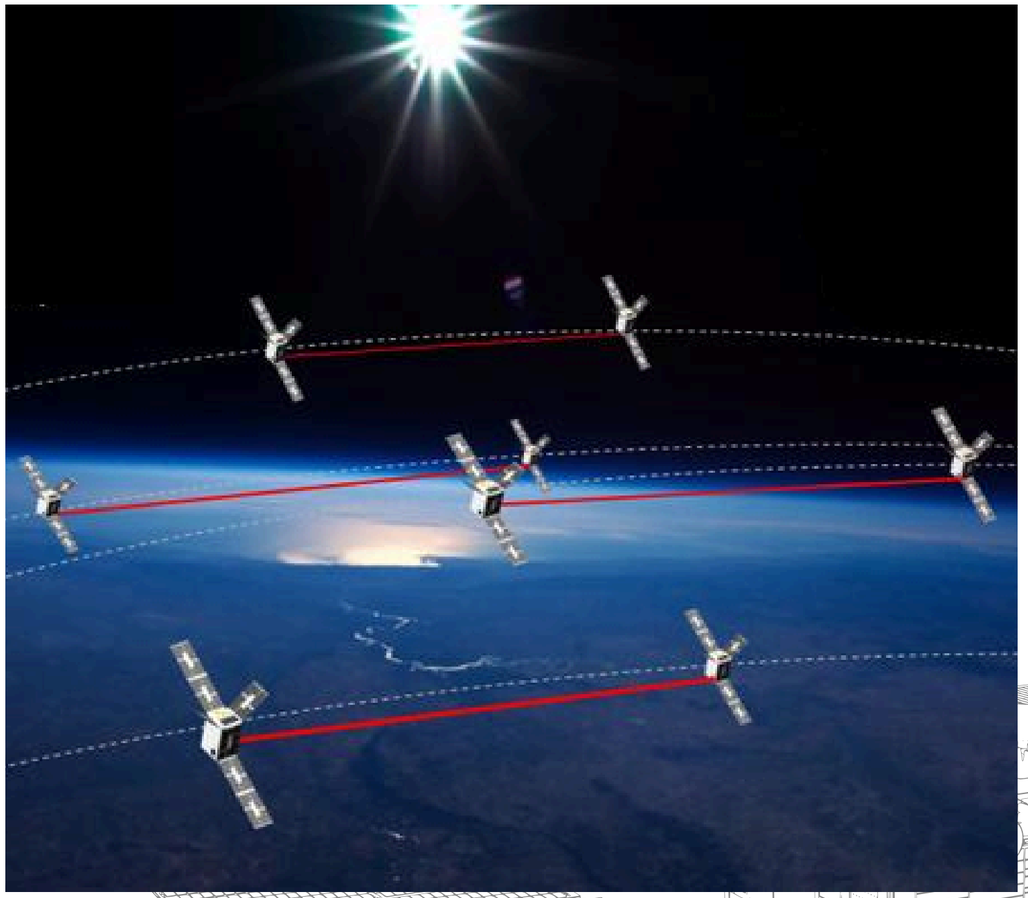
Introduction

This project proposes a decentralized Federated Learning framework for multi-satellite swarms to autonomously detect subsystem anomalies, generate real-time health scores, and predict subsystem failures through cluster-level weight collaboration.

Abstract

This project proposes a **decentralized Federated Learning framework** designed for autonomous anomaly detection in multi-satellite systems.

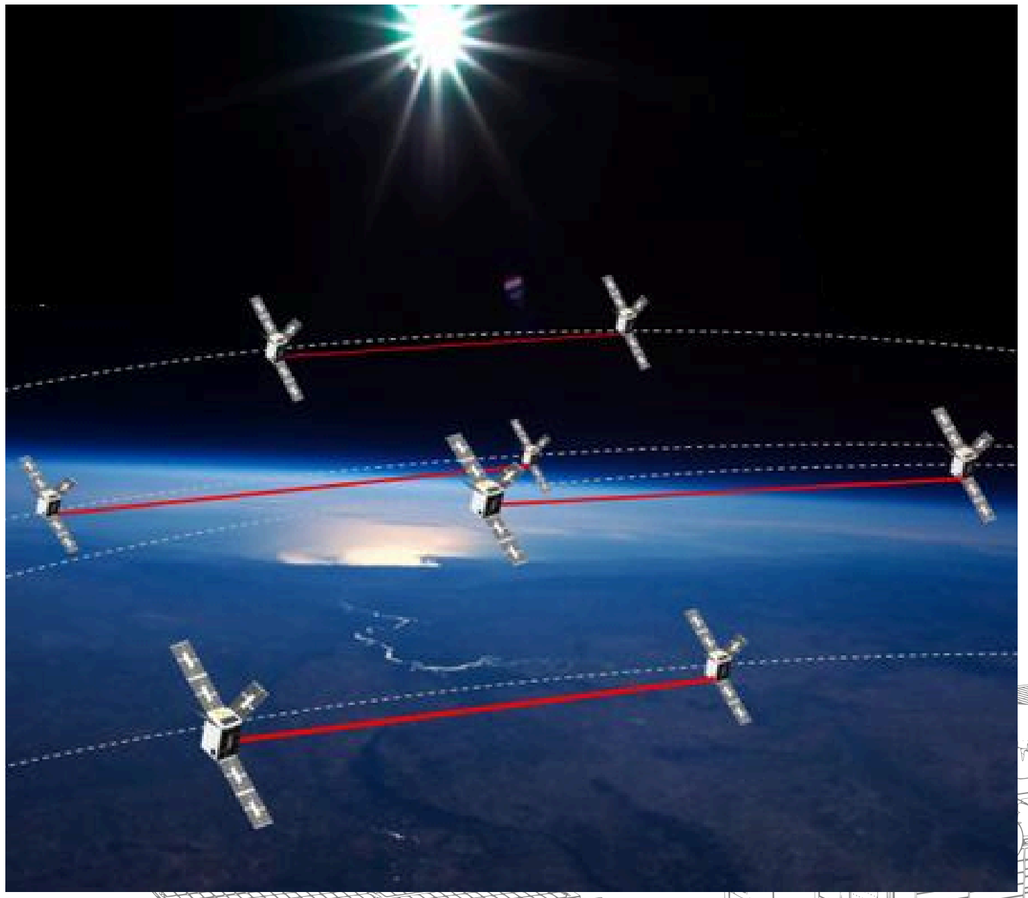
By shifting intelligence to the orbital edge, individual satellites process high-dimensional telemetry locally to generate real-time health indices, sharing only optimized model updates to iteratively improve a **collective global model**.



Abstract

To maintain system integrity, the framework employs a **health-aware leader selection process** and robust statistical aggregation to identify and discard weight updates from faulty or anomalous nodes.

This architecture is completed by an **automated alerting system** that provides **predictive failure notifications** to ground stations, ensuring proactive mission management and reduction in downlink requirements.



Problem Statement

Federated Learning Architecture for Health Monitoring and Anomaly Detection in Multi Satellite Systems

This project proposes a decentralized Federated Learning framework for multi-satellite swarms to autonomously detect subsystem anomalies, generate real-time health scores, and predict subsystem failures through cluster-level weight collaboration. The architecture utilizes robust aggregation to identify and remove faulty node updates while providing automated ground alerts for proactive failure management without ground-station dependency.

Suggestions

Feasibility of Predictive Maintenance

True predictive maintenance requires long-term hardware degradation data, fleet-scale data from thousands of nodes, and industrial-grade validation are not feasible.

Scope narrowed to anomaly detection & health monitoring. Problem statement updated to: Federated Learning Architecture for Health Monitoring and Anomaly Detection in Multi-Satellite Systems.

Types of Anomalies to Detect

Original scope was limited to hardware anomalies, which occur infrequently in small satellites and would result in a sparse training signal.

Expanded to include software faults, payload anomalies, communication gaps, and rare nominal events triggered by telecommands that are fully captured in the ESA-ADB Mission 1 annotation scheme.

Uniform Feature Representation

Having ZOH channels with 1 feature and aggregate channels with 5 features creates non-uniform input tensors which are incompatible with standard GNN architectures like GDN.

Reprocessed all 76 channels to produce 5 uniform features (mean, max, min, std, count) per 30 s window. ZOH channels produce degenerate values (std=0, count=1) which are themselves informative signals.

Analysis and Critique of Research

Adaptive FL for Satellite Telemetry by Atefrad & Karami (2024)

Strengths: Uses ESA-ADB; demonstrates FL reduces comms to 1.8 MB/round; LSTM + FedAvg + fine-tuning achieves F0.5 = 0.86.
Gaps: Simulates FL by splitting time periods, not channels, so not physically realistic. Uses Mission 2 lightweight (11 ch only). No distinction between anomalies and rare nominal events. No per-channel timestamp handling.
Our Improvement: Channel-partitioned cross-satellite federation on Mission 1 (76 ch). Three-class label scheme. Per-channel timestamps.

GDN: Graph Deviation Network by Deng & Hooi (2021)

Strengths: Learns inter-sensor graph via attention; detects anomalies as deviations from learned relationships; outperforms LSTM on SWaT and MSDS benchmarks.
Gaps: Designed for single-node operation. Not evaluated on satellite telemetry. No federated extension. Requires careful graph initialization.
Our Extension: First application of GDN in federated satellite setting. Per-satellite graph over local channel subsets. FedProx for non-IID channel heterogeneity.

Relevance and Real World Application

ESA Operations

Directly addresses ESA's stated goal of autonomous onboard anomaly detection. ESA-ADB benchmark aligns with real operational constraints.

NASA Deep Space

NASA plans 10–100 satellite swarms for deep space. Communication delays make ground-based monitoring infeasible; onboard FL is the only viable path.

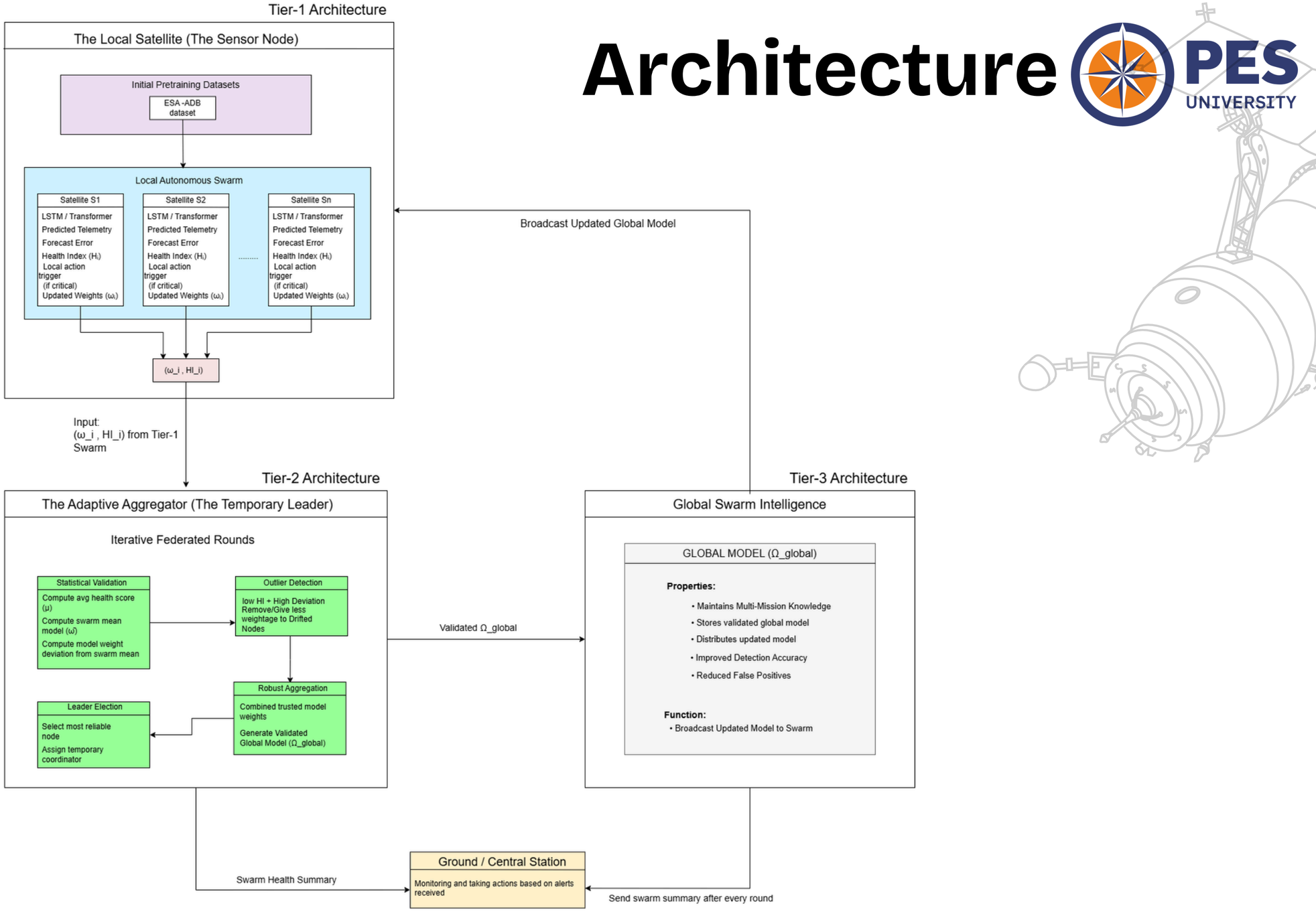
Commercial Constellations

SpaceX Starlink, OneWeb, and Amazon Kuiper deploy hundreds of satellites. Federated health monitoring scales where centralised telemetry cannot.

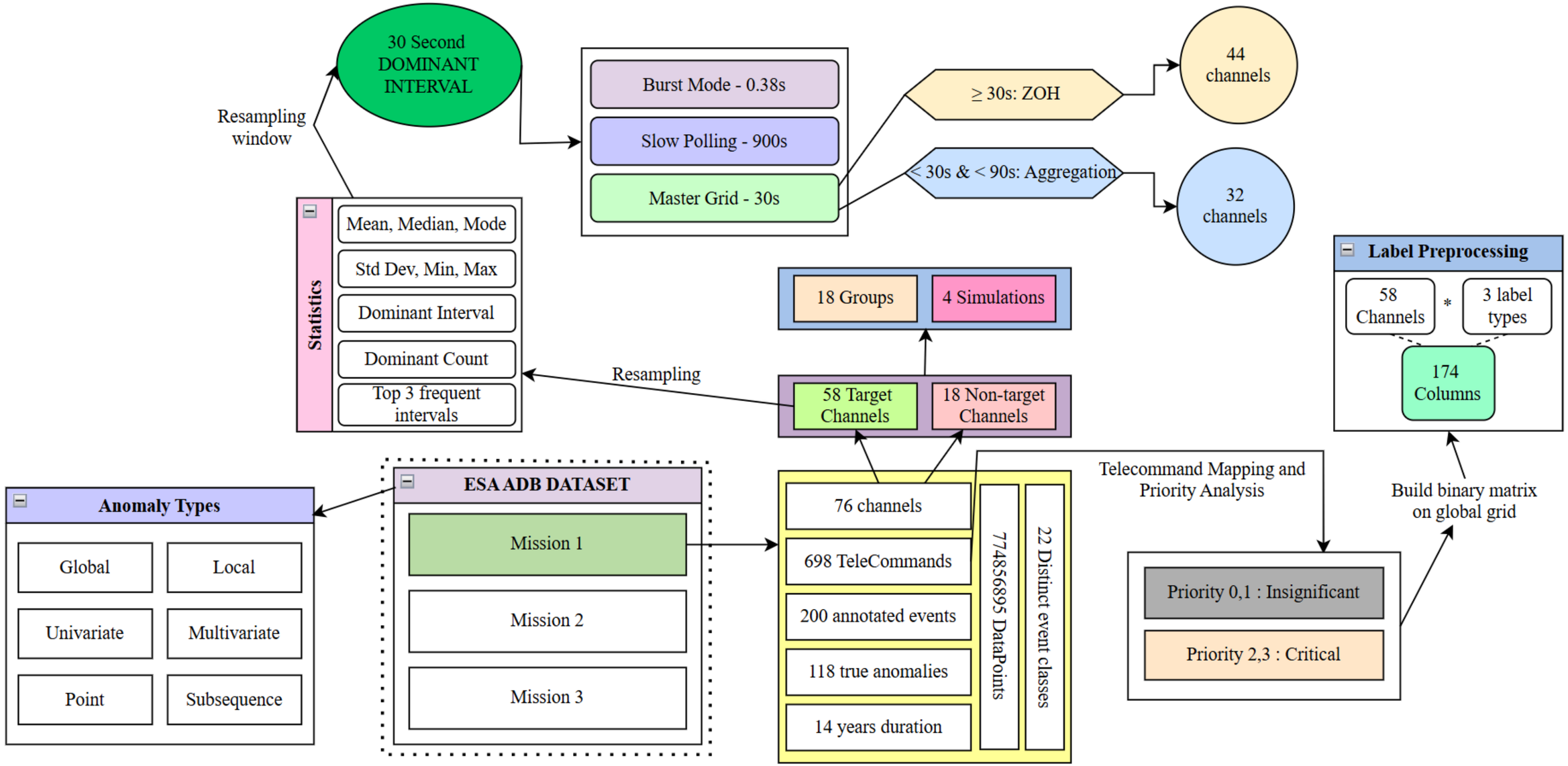
Bandwidth Economics

Target 90% downlink reduction: transmitting model weights (~MBs/round) vs raw telemetry (~GBs/day) reduces ground infrastructure cost significantly.

Architecture



Data Processing & Analysis



Design Methodology

Data Preprocessing

- ESA-ADB Mission 1 Dataset (76 channels, 14 years)
- Pandas / NumPy resampling & feature extraction
- Scikit-Learn Z-score normalisation per phase
- Custom 5-feature aggregation (mean/max/min/std/count)
- Per-channel binary label annotation (3-class scheme)

Model Architecture

- Graph Deviation Network (GDN) Deng & Hooi 2021
- PyTorch + PyTorch Geometric
- Temporal sliding windows: 256 steps × 30 s = 128 min
- Per-satellite sensor graph learned via attention
- Reconstruction error → normalised Health Index (0–1)

Federated Learning

- Flower (flwr) framework : client/server protocol
- FedProx (μ=0.01) proximal regularisation
- Trimmed Mean aggregation z-score outlier filtering
- Health-based dynamic leader election every N rounds
- tc_fired binary feature: priority 2/3 TC context

Safety & Alerting

- NPDT thresholding : dynamic anomaly boundary
- Linear Regression RUL predictor on HI history
- Autonomous FDIR safe-mode lookup table
- Emergency ground alerts when RUL < n-day threshold
- Swarm Status Packet : compact downlink summary

Dataset

Table 1. **Summary statistics of missions included in ESA-ADB.**

	Mission1	Mission2	Both missions
Channels	**76**	**100**	**176**
Target / Non-target	58 / 18	47 / 53	105 / 71
Channel groups	18	29	47
Subsystems	4	5	6*
Telecommands	**698**	**123**	**821**
Priority 0/1/2/3	345 / 323 / 19 / 11	0 / 0 / 119 / 4	345 / 323 / 138 / 15
Total executions	1,594,722	1,918,002	3,512,724
Data points	**774,856,895**	**776,734,364**	**1,551,591,259**
Duration (anonymised)	14 years	3.5 years	17.5 years
Compressed size [GB]	3.51	3.81	7.32
Annotated points [%]	1.80	0.58	1.19
Annotated events	**200**	**644**	**844**
Anomalies	118	31	148
Rare nominal events	78	613	690
Communication gaps	4	0	4
Univariate / Multivariate	32 / 164	9 / 635	41 / 799
Global / Local	113 / 83	585 / 59	698 / 142
Point / Subsequence	12 / 184	0 / 644	12 / 828
Distinct event classes	**22**	**32****	**54**

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Channel Metrics & Sampling Rates

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channel_name	total_intervals	mean_s	median_s	mode_s	std_dev_s	min_s	max_s	dominant_interval	dominant_count	dominant_pct	top1_interval	top1_pct	top2_interval	top2_pct	top3_interval	top3_pct	source_file
Group1/channel_1.csv	10513335	42.0275	3.0	3.0	66.9384	0.003	102573.0	3.0	5430647	0.5165	3.0	0.5165	90.0	0.44530000000000000	2.99	0.018600000000000000	interval_results/Group1/channel_1.txt
Group1/channel_2.csv	10513335	42.0275	3.0	3.0	66.9384	0.003	102573.0	3.0	5430647	0.5165	3.0	0.5165	90.0	0.44530000000000000	2.99	0.018600000000000000	interval_results/Group1/channel_2.txt
Group1/channel_3.csv	10513335	42.0275	3.0	3.0	66.9384	0.003	102573.0	3.0	5430647	0.5165	3.0	0.5165	90.0	0.44530000000000000	2.99	0.018600000000000000	interval_results/Group1/channel_3.txt
Group2/channel_4.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_4.txt
Group2/channel_5.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_5.txt
Group2/channel_6.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_6.txt
Group2/channel_7.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_7.txt
Group2/channel_8.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_8.txt
Group2/channel_9.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_9.txt
Group2/channel_10.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_10.txt
Group2/channel_11.csv	3589375	123.0993	167.25	180.0	83.7687	0.003	21420.0	180.0	1632271	0.4548	180.0	0.4548	0.75	0.1264	47.25	0.0655	interval_results/Group2/channel_11.txt
Group3/channel_12.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_12.txt
Group3/channel_13.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_13.txt
Group4/channel_14.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group4/channel_14.txt
Group7/channel_15.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group7/channel_15.txt
Group6/channel_16.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group6/channel_16.txt
Group5/channel_17.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_17.txt
Group5/channel_18.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_18.txt
Group3/channel_19.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_19.txt
Group3/channel_20.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_20.txt
Group4/channel_21.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group4/channel_21.txt
Group7/channel_22.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group7/channel_22.txt
Group7/channel_23.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group7/channel_23.txt
Group6/channel_24.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group6/channel_24.txt
Group5/channel_25.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_25.txt
Group5/channel_26.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_26.txt
Group3/channel_27.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_27.txt
Group3/channel_28.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group3/channel_28.txt
Group4/channel_29.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group4/channel_29.txt
Group4/channel_30.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group4/channel_30.txt
Group7/channel_31.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group7/channel_31.txt
Group6/channel_32.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group6/channel_32.txt
Group6/channel_33.csv	5272892	83.7964	90.0	90.0	29.7491	0.36	21330.0	90.0	4860121	0.9217	90.0	0.9217	3.0	0.068	0.38	0.0078000000000000000	interval_results/Group6/channel_33.txt
Group5/channel_34.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_34.txt
Group5/channel_35.csv	14258505	30.9885	0.375	0.38	44.0743	0.003	21330.0	0.38	8328077	0.5841	0.38	0.5841	90.0	0.3382	0.37	0.0466	interval_results/Group5/channel_35.txt

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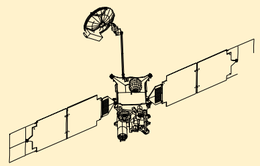
Channel Metrics & Sampling Rates for aggregate channels

Total channels	ZOH (1 feature)	Aggregate (5 features)	IN LIST
76	44	32	32 / 32

Aggregate channels

CHANNELS	GROUP	MODE INTERVAL	DOMINANT %	TRIGGER RULE	IN LIST	TARGET?
ch 1, 2, 3	Group 1	3s	51.65%	pct < 90% AND interval < 30s	yes ✓	non-target
ch 4–11	Group 2	180s	45.48%	pct < 90%	yes ✓	non-target
ch 12, 13	Group 3	0.38s	58.41%	pct < 90% AND interval < 30s	yes ✓	target
ch 17, 18, 19, 20	Groups 3,5	0.38s	58.41%	pct < 90% AND interval < 30s	yes ✓	target
ch 25, 26, 27, 28	Groups 3,5	0.38s	58.41%	pct < 90% AND interval < 30s	yes ✓	target
ch 34, 35, 36, 37	Groups 3,5	0.38s	58.41%	pct < 90% AND interval < 30s	yes ✓	target
ch 64, 65, 66	Groups 15,16	180s	49.53%	pct < 90%	yes ✓	target
ch 67, 68, 69	Group 17	180s	49.53%	pct < 90%	yes ✓	non-target
ch 74	Group 18	30s	70.83%	pct < 90%	yes ✓	target

ZOH Channel (1 feature) — 44 channels
dominant_pct >= 90% AND
dominant_interval >= 30s



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Channel Metrics & Sampling Rates for ZOH channels

ZOH channels — validated

CHANNELS	GROUP	MODE INTERVAL	DOMINANT %	BOTH RULES PASS?	TARGET?
ch 14, 15, 16	Groups 4,6,7	90s	92.17%	✓ pct≥90% AND ≥30s	target
ch 21, 22, 23, 24	Groups 4,6,7	90s	92.17%	✓ pct≥90% AND ≥30s	target
ch 29, 30, 31, 32, 33	Groups 4,6,7	90s	92.17%	✓ pct≥90% AND ≥30s	target
ch 38, 39, 40	Groups 4,6,7	90s	92.17%	✓ pct≥90% AND ≥30s	target
ch 41–46	Group 8	30s	90.28%	✓ pct≥90% AND ≥30s	target
ch 47, 48, 49	Group 9	30s	99.81%	✓ pct≥90% AND ≥30s	target
ch 50, 51, 52	Group 10	90s	97.88%	✓ pct≥90% AND ≥30s	target
ch 53, 54, 55, 56	Groups 11,12	30s	99.85%	✓ pct≥90% AND ≥30s	non-target
ch 57, 58, 59, 60	Group 13	30s	99.85%	✓ pct≥90% AND ≥30s	target
ch 61, 62, 63	Group 14	900s	99.95%	✓ pct≥90% AND ≥30s	target
ch 70, 71, 72, 73	Group 18	30s	~99.93%	✓ pct≥90% AND ≥30s	target
ch 75, 76	Group 18	30s	99.94%	✓ pct≥90% AND ≥30s	target

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Analysis of Anomaly Durations

anomaly_durations

channel	start_time	end_time	duration_seconds
channel_12	2004-12-01 20:42:15.429000+00:00	2004-12-08 22:55:45.429000+00:00	612810.0
channel_13	2004-12-01 20:42:15.429000+00:00	2004-12-08 22:55:45.429000+00:00	612810.0
channel_14	2004-12-01 20:43:45.429000+00:00	2004-12-02 02:57:15.429000+00:00	22410.0
channel_15	2004-12-01 20:45:00.429000+00:00	2004-12-02 02:58:45.429000+00:00	22425.0
channel_16	2004-12-01 20:43:45.429000+00:00	2004-12-16 16:52:30.429000+00:00	1282125.0
channel_17	2004-12-01 20:45:00.429000+00:00	2004-12-08 22:55:45.429000+00:00	612645.0
channel_18	2004-12-01 20:45:00.429000+00:00	2004-12-08 22:55:45.429000+00:00	612645.0
channel_19	2004-12-01 20:47:03.429000+00:00	2004-12-08 22:55:45.429000+00:00	612522.0
channel_20	2004-12-01 20:45:00.429000+00:00	2004-12-08 22:55:45.429000+00:00	612645.0
channel_21	2004-12-01 20:45:00.429000+00:00	2004-12-02 02:57:15.429000+00:00	22335.0
channel_22	2004-12-01 20:45:00.429000+00:00	2004-12-02 02:57:15.429000+00:00	22335.0
channel_23	2004-12-01 20:45:00.429000+00:00	2004-12-02 02:58:45.429000+00:00	22425.0
channel_24	2004-12-01 20:43:45.429000+00:00	2004-12-16 16:52:30.429000+00:00	1282125.0
channel_25	2004-12-01 20:43:45.429000+00:00	2004-12-08 22:55:45.429000+00:00	612720.0

anomaly_types

ID	Class	Subclass	Category	Dimensionality	Locality	Length
id_1	class_6	subclass_1	Rare Event	Multivariate	Global	Subsequence
id_2	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_3	class_7	subclass_1	Anomaly	Multivariate	Global	Subsequence
id_4	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_5	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_6	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_7	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_8	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_9	class_7	subclass_1	Anomaly	Multivariate	Local	Subsequence
id_10	class_9	subclass_1	Anomaly	Multivariate	Global	Subsequence
id_11	class_11	unknown	Anomaly	Multivariate	Local	Subsequence
id_12	class_11	unknown	Anomaly	Multivariate	Global	Subsequence
id_13	class_11	unknown	Anomaly	Multivariate	Global	Subsequence
id_14	class_4	subclass_1	Rare Event	Multivariate	Global	Subsequence

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Telecommand (TC) Mapping & Priority Analysis

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telecommands

Telecommand	Priority
telecommand_1	1
telecommand_2	1
telecommand_3	1
telecommand_4	1
telecommand_5	1
telecommand_6	1
telecommand_7	1
telecommand_8	1
telecommand_9	1
telecommand_10	1
telecommand_11	1
telecommand_12	1
telecommand_13	1
telecommand_14	1
telecommand_15	1

telecommand_1

	telecommand_1
2001-06-12 22:27:20.292	1
2005-01-23 00:45:00.000	1
2005-01-23 01:38:26.496	1
2005-04-01 07:28:24.984	1
2005-05-10 23:52:51.177	1
2005-09-02 18:24:13.740	1
2005-09-28 10:19:24.249	1
2005-10-22 23:02:28.104	1
2007-04-14 13:24:17.235	1
2007-04-14 20:01:15.039	1
2009-05-26 11:57:34.428	1
2009-05-26 12:00:46.998	1
2009-05-30 10:21:43.518	1
2009-05-30 10:27:42.915	1
2010-12-17 04:24:32.016	1

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DATA RESAMPLING AND TRAINING STATS

```
datetime,channel_1_mean,channel_1_max,channel_1_min,channel_1_std,channel_1_count
2000-01-01 00:00:00+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:00:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:01:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:01:30+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:02:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:02:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:03:00+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:03:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:04:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:04:30+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:05:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:05:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:06:00+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:06:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:07:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:07:30+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:08:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:08:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:09:00+00:00,0.13791034,0.13791034,0.13791034,0.0,1
2000-01-01 00:09:30+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:10:00+00:00,0.13791034,0.13791034,0.13791034,0.0,0
2000-01-01 00:10:30+00:00,0.13791034,0.13791034,0.13791034,0.0,1
```

Users > Shreyas > Downloads > Capstone > Datasets > ESA > ESA-Mission1 > interval_results > Group1 > channel_1.txt

```
================================================================================
  Channel       : Group1/channel_1.csv
================================================================================
  Total rows    : 10,513,336
  Valid rows    : 10,513,336  |  Dropped (NaT): 0
  Total intervals: 10,513,335
--------------------------------------------------------------------------------
  INTERVAL STATISTICS  (seconds between consecutive timestamps)
--------------------------------------------------------------------------------
  Mean              : 42.0275 s
  Median            : 3.0000 s
  Mode              : 3.00 s
  Std Dev           : 66.9384 s
  Min               : 0.0030 s
  Max               : 102573.0000 s
--------------------------------------------------------------------------------
  Dominant interval : 3.00 s
  Dominant count    : 5,430,647  (51.65% of all intervals)
--------------------------------------------------------------------------------
  TOP 3 MOST FREQUENT INTERVALS:
    #1  3.00 s  →  5,430,647 occurrences  (51.65%)
    #2  90.00 s  →  4,681,837 occurrences  (44.53%)
    #3  2.99 s  →  195,329 occurrences  (1.86%)
--------------------------------------------------------------------------------
  FIRST 5 INTERVALS (s) :
    [90.0, 90.0, 90.0, 90.0, 90.0]

  LAST 5 INTERVALS (s)  :
    [90.0, 90.0, 90.0, 90.0, 90.0]
================================================================================
```

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Users > Shreyas > Downloads > Capstone > Datasets > ESA > ESA-Mission1 > interval_results > Group5 > channel_34.txt

```
================================================================
  Channel       : Group5/channel_34.csv
================================================================
  Total rows    : 14,258,506
  Valid rows    : 14,258,506  |  Dropped (NaT): 0
  Total intervals: 14,258,505
----------------------------------------------------------------
  INTERVAL STATISTICS  (seconds between consecutive timestamps)
----------------------------------------------------------------
  Mean              : 30.9885 s
  Median            : 0.3750 s
  Mode              : 0.38 s
  Std Dev           : 44.0743 s
  Min               : 0.0030 s
  Max               : 21330.0000 s
----------------------------------------------------------------
  Dominant interval : 0.38 s
  Dominant count    : 8,328,077  (58.41% of all intervals)
----------------------------------------------------------------
  TOP 3 MOST FREQUENT INTERVALS:
    #1  0.38 s  →  8,328,077 occurrences  (58.41%)
    #2  90.00 s  →  4,822,681 occurrences  (33.82%)
    #3  0.37 s  →  664,016 occurrences  (4.66%)
----------------------------------------------------------------
  FIRST 5 INTERVALS (s) :
    [90.0, 90.0, 90.0, 90.0, 90.0]

  LAST 5 INTERVALS (s)  :
    [90.0, 90.0, 90.0, 90.0, 90.0]
================================================================
```

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```
================================================================
  GDN — Graph Deviation Network — ESA Satellite Telemetry
  Channels: 41–46  |  Feature: channel value (1/channel)
================================================================

[1/5] Loading telemetry data …
    Aligned timesteps : 14,728,320  (122736.0 hours)
    Data matrix shape : (14728320, 6)  [T × 6 channels]

[2/5] Loading labels and building windows …
    Total windows     : 215  (stride=68503)
    Nominal windows   : 199  → training
    Anomaly windows   : 16   → evaluation only
    Train set (nominal only): 159 windows
    Val set  (nom+anom)     : 56 windows  (40 nom / 16 anom)

[3/5] GDN ready — 145,210 trainable parameters
      n_nodes=6  embed_dim=64  gru_hidden=64  feat/node=1

[4/5] Training on 199 nominal windows for 15 epochs …
      Task: predict next timestep | Anomaly = high deviation

  Epoch 01/15 | Loss: 0.8012 | Batches: 5 | Time: 0.8s | 4.8 bpt
  Epoch 02/15 | Loss: 0.6834 | Batches: 5 | Time: 0.9s | 4.8 bpt
  Epoch 03/15 | Loss: 0.5721 | Batches: 5 | Time: 1.2s | 4.8 bpt
  Epoch 04/15 | Loss: 0.4893 | Batches: 5 | Time: 1.3s | 4.8 bpt
  Epoch 05/15 | Loss: 0.4312 | Batches: 5 | Time: 1.1s | 4.8 bpt
  Epoch 06/15 | Loss: 0.3971 | Batches: 5 | Time: 1.1s | 4.8 bpt
  Epoch 07/15 | Loss: 0.3754 | Batches: 5 | Time: 0.9s | 4.8 bpt
  Epoch 08/15 | Loss: 0.3601 | Batches: 5 | Time: 1.3s | 4.8 bpt
  Epoch 09/15 | Loss: 0.3489 | Batches: 5 | Time: 1.3s | 4.8 bpt
  Epoch 10/15 | Loss: 0.3412 | Batches: 5 | Time: 1.2s | 4.8 bpt
  Epoch 11/15 | Loss: 0.3358 | Batches: 5 | Time: 1.0s | 4.8 bpt
  Epoch 12/15 | Loss: 0.3319 | Batches: 5 | Time: 0.9s | 4.8 bpt
  Epoch 13/15 | Loss: 0.3289 | Batches: 5 | Time: 0.9s | 4.8 bpt
  Epoch 14/15 | Loss: 0.3268 | Batches: 5 | Time: 1.0s | 4.8 bpt
  Epoch 15/15 | Loss: 0.3102 | Batches: 5 | Time: 0.9s | 4.8 bpt

  Training complete ✓

  Learned sensor attention matrix (6×6):
  Rows = which channel is PREDICTED, Cols = which channel it ATTENDS TO
            41    42    43    44    45    46
  ------------------------------------------------
  ch_41  →   0.314  0.152  0.081  0.221  0.091  0.141
  ch_42  →   0.142  0.421  0.051  0.162  0.153  0.071
  ch_43  →   0.061  0.092  0.512  0.081  0.114  0.140
  ch_44  →   0.210  0.131  0.072  0.352  0.101  0.134
  ch_45  →   0.111  0.082  0.183  0.134  0.412  0.078
  ch_46  →   0.152  0.123  0.141  0.165  0.081  0.338
```

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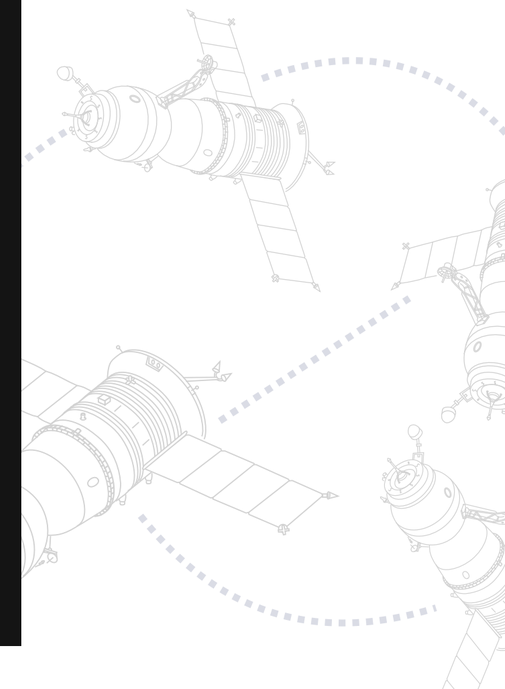

Data Pre-Processing Results

The primary objective is to create a **single, globally aligned 30-second time grid** (spanning years 2000 to 2014 in UTC) to generate three core data structures:

- **Resampled Telemetry Features:** Cleaned sensor data per channel, optimized for Deep Learning.
- **Telecommand Firing Matrix:** A binary matrix of commands aligned to the global grid.
- **Label Master Matrix:** Ground truth indicators for anomalies, rare events, and communication gaps.

Critical Alignment: This synchronization ensures that features, commands, and labels share identical timestamps, allowing them to be concatenated into a fixed-width training tensor.

Technologies Used

Tools (libraries/modules)

- pandas — read/write CSVs, parse datetimes, compute time deltas, resample onto fixed grids, aggregate window stats, fill missing values, merge/join tables.
- numpy — build fast binary label arrays, compute nearest-grid index via vector math, and do efficient boolean/array operations.
- pathlib (Path) — robust path handling, recursive file discovery (rglob), and writing outputs to structured folders.
- os — filesystem operations (join paths, existence checks, create directories).
- re (regex) — extract IDs/channel numbers, parse metrics out of .txt reports

Technologies Used

Technologies / data formats — with how they were used

- Python scripts — orchestration + analysis pipelines end-to-end.

CSV datasets — primary storage for time series channels, labels, telecommands, and produced feature/merge tables.

Conclusion

✓

Complete Preprocessing Pipeline

All 76 channels resampled to a uniform 5-feature 30s grid. 14,728,321-row master_dataset.csv built with labels and telecommand context. Ready for model training.

✓

Rigorous Label Annotation

Per-channel three-class labels (anomaly / rare event / comm gap) built from actual ESA-ADB timestamps. 248 multi-window events and 12 point anomalies correctly handled.

✓

Benchmark-Aligned Evaluation Setup

5-phase chronological train/val split following ESA-ADB protocol. Per-phase normalisers. 256-step windows with stride 64 generated. Directly comparable to published leaderboard.

→

Next: GDN Model + Federated Loop

Graph Deviation Network per satellite node. FedProx + Trimmed Mean aggregation. Health-based leader election. Evaluation against ESA-ADB event-wise F0.5 metric.

Individual Contributions

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Shreyas

- Channel interval analysis and computed mean/median/mode/std of all 76 channels
- ZOH vs aggregate classification logic (dominant_pct ≥ 90%, interval ≥ 30s rule)
- Identified bimodal sampling in 26 channels and extreme gap anomalies in Groups 14 & 18
- Led architecture design: 3-tier FL system, 6-phase operational workflow

Tulika

- Implemented uniform 5-feature resampling pipeline for all 76 channels
- Built labels_master.csv — per-channel label stamping with per-channel timestamps
- Handled 248 multi-window events and 12 point anomaly snap-to-grid cases
- Validated slot counts against expected values from annotation durations

Meghana

- Built telecommands_master.csv and filtered to 30 priority-2/3 TCs, snapped to 30s grid
- Merged all 76 channels + labels + tc_fired into master_dataset.csv (381 columns)
- Implemented 5-phase benchmark train/val split and per-phase Z-score normalisation
- Generated 256-timestep sliding windows with stride 64 for model input

Vijay

- Designed all UML diagrams: Master Class Diagram, ER Diagram, Use Case Diagram
- Designed External Interface Diagram and federated communication protocol
- Literature review: GDN (Deng & Hooi), FedProx (Li et al.), ESA-ADB benchmark
- Initiated GDN model implementation in PyTorch Geometric [in progress]

Implementation Stats

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30%

✓ DONE — Channel Interval Analysis

Computed inter-sample interval statistics for all 76 channels. Classified 44 as ZOH (dominant_pct ≥ 90%, interval ≥ 30s) and 32 as aggregate channels. Bimodal distributions identified in 26 channels.

✓ DONE — Uniform 5-Feature Resampling

All 76 channels resampled to 30s grid with 5 features (mean, max, min, std, count). ZOH channels produce std=0, count=1 — informative as mode signals. ResampledData/ folder fully populated.

✓ DONE — Labels Master Construction

Per-channel binary label files built for all 58 target channels. Three-class scheme: is_anomaly, is_rare_event, is_comm_gap. 248 multi-window events handled correctly. 12 point anomalies snapped to 30s grid slots.

✓ DONE — Telecommands Master

telecommands.csv filtered to 30 priority-2/3 TCs. Individual TC CSV files read and firing timestamps snapped to nearest 30s grid slot. Single binary column tc_fired built on master grid.

✓ DONE — Master Dataset Merge

All 76 × 5 = 380 feature columns joined with labels and tc_fired on datetime index. Final master_dataset.csv: 381 feature columns + 3 label columns on 14,728,321 rows (2000-01-01 → 2014-01-01).

✓ DONE — Sliding Windows & Phase Splits

5-phase chronological train/val split following ESA-ADB benchmark protocol. Z-score normalisation fitted on nominal training data only. 256-timestep windows with stride 64 generated. Ready for model input.

IN PROG — GDN Model Implementation

Graph Deviation Network implementation in PyTorch Geometric. Per-satellite sensor graph construction. FedProx training loop with Flower framework. Training not yet started.

PENDING — Federated Training & Evaluation

FedProx + Trimmed Mean aggregation loop. Dynamic leader election. FDIR alerting. Evaluation against ESA-ADB event-wise F0.5 metric.

Project Timeline

Phase I: Foundation | Phase II: Implementation (Current) | Phase III: Validation & Publication

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CAPSTONE -II

Thank you

Review - 3